Deep Learning–Based Multi-Class Skin Lesion Classification Using EfficientNet

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***Abstract*—Early detection of skin cancer significantly improves patient outcomes, but access to expert dermatologists remains limited in many regions. We present SkinCare AI, a web-based tool that classifies dermoscopic images into eight lesion categories. Our approach compares a baseline CNN with an EfficientNet-based architecture using transfer learning and data augmentation to improve robustness. We trained and evaluated models on the ISIC 2019 dataset, measuring accuracy, precision, recall, and F1-score. Results show the EfficientNet model outperforms the baseline across all metrics. The trained classifiers are deployed in a secure web platform with user authentication, analysis history, PDF report generation, and Grad-CAM visualizations. All outputs include disclaimers stating this tool is for research purposes only, not clinical diagnosis. Our work demonstrates that efficient, interpretable deep learning models can support scalable dermatological screening.**

***Index Terms*—Skin lesion classification, EfficientNet, deep learning, ISIC 2019, transfer learning, explainable AI, Grad-CAM**

I. Introduction

Skin cancer is among the most common cancers worldwide, and early detection greatly improves survival rates [1]. While dermoscopy is a key diagnostic tool, skilled interpretation requires specialized training that is often unavailable in underserved areas. This gap motivates the development of automated screening tools for research and preliminary assessment.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have shown impressive results in medical imaging [2]. Transfer learning allows models pretrained on large image datasets to perform well on medical tasks with limited labeled data [3]. Among modern architectures, EfficientNet balances accuracy and computational cost through compound scaling [4].

However, the lack of model interpretability hinders clinical adoption [5]. Techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) address this by highlighting image regions that influence predictions [6].

This paper presents SkinCare AI, a web-based research prototype that combines an EfficientNet-based skin lesion classifier with explainability features. We evaluate our approach on the ISIC 2019 dataset for eight-class lesion classification. Our main contributions are:

- An EfficientNetB0-based multi-class skin lesion classifier
- Systematic comparison with a baseline CNN
- Comprehensive evaluation using standard classification metrics
- Integration of Grad-CAM for visual model interpretation

II. Related Work

Early work on automated skin lesion analysis used established CNN architectures such as VGG [7] and ResNet [8]. More recent studies have explored efficient architectures like DenseNet [9] and EfficientNet [4], achieving better results on benchmarks including ISIC.

ISIC challenges have highlighted key obstacles in skin lesion classification: class imbalance and visual similarity between lesion types [10]. Researchers have addressed these through ensemble methods, weighted loss functions, and aggressive augmentation [11].

Model interpretability has become essential in medical AI. Grad-CAM and similar visualization methods help verify that CNNs focus on clinically relevant features rather than artifacts [6]. Attention mechanisms have also shown promise for highlighting discriminative features [15].

Advanced augmentation techniques like mixup and cutmix help address class imbalance [16], and incorporating patient metadata can further improve classification [17].

Unlike complex ensemble approaches, our work focuses on building a transparent, reproducible baseline using EfficientNetB0 with explainability features.

III. Dataset Description

We use the ISIC 2019 dataset [12], which contains 25,331 dermoscopic images across eight diagnostic categories: Actinic Keratoses (AK), Basal Cell Carcinoma (BCC), Benign Keratosis (BKL), Dermatofibroma (DF), Melanoma (MEL), Melanocytic Nevi (NV), Squamous Cell Carcinoma (SCC), and Vascular Lesions (VASC).

The dataset exhibits significant class imbalance: NV is the majority class (12,875 samples) while DF is the minority (239 samples). Images were resized to 224×224 pixels and normalized using ImageNet statistics. We split the data into training (70%), validation (15%), and test (15%) sets using stratified sampling. Fig. 1 shows sample images from each category.

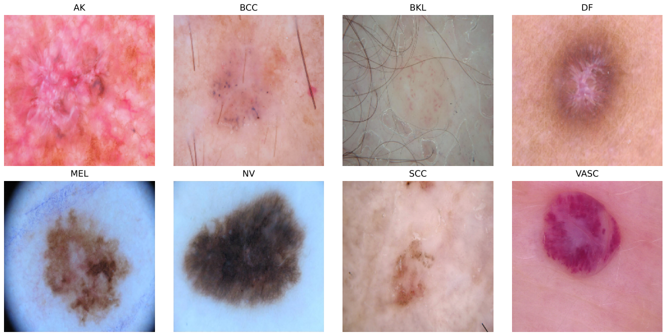

Fig. 1. Sample dermoscopic images from the ISIC 2019 dataset representing eight lesion categories: AK, BCC, BKL, DF, MEL, NV, SCC, and VASC.

IV. Methodology

A. Evaluation Metrics

Classification performance was assessed using standard metrics. For multi-class classification problems, these metrics are computed as follows:

Accuracy measures the proportion of correctly classified instances among all instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

Precision quantifies the proportion of correctly predicted positive observations to total predicted positive observations:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall (Sensitivity) measures the proportion of correctly predicted positive observations to all actual positive observations:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

F1-Score provides the harmonic mean of precision and recall, offering a balanced measure particularly useful for imbalanced datasets:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \tag{4}$$

For multi-class scenarios, both macro-averaged and weighted-averaged metrics are computed to provide comprehensive performance assessment across all classes.

B. Model Architecture

We use EfficientNetB0 pretrained on ImageNet as our backbone [4]. EfficientNet employs compound scaling to uniformly scale network width, depth, and resolution. We replace the original classification head with a custom classifier: global average pooling, batch normalization, dropout (rate=0.3), a 256-neuron dense layer with ReLU activation, additional batch normalization, dropout (rate=0.2), and a softmax output layer for eight-class prediction. The model has approximately 4.38 million total parameters, with 329,992 trainable parameters in the classifier head.

Fig. 2 shows the complete architecture from input to classification output.

C. Baseline CNN

For comparison, we implement a baseline CNN with three convolutional blocks (32, 64, and 128 filters) using 3×3 kernels, batch normalization, ReLU activation, and 2×2 max-pooling. The convolutional layers are followed by global average pooling, a 256-neuron dense layer with dropout, and an eight-class softmax output.

D. Training Strategy

Both models are trained using categorical cross-entropy loss with the Adam optimizer [13]. We use an initial learning rate of 1×10^{-3} with reduction on plateau (factor=0.3, patience=2). Early stopping (patience=5) prevents overfitting. Data augmentation includes horizontal flips, rotations (± 10), zoom ($\pm 10\%$), and contrast adjustments. Training uses mixed precision (float16) for efficiency and runs for 8 epochs, saving the best checkpoint based on validation accuracy.

E. Explainability with Grad-CAM

Grad-CAM produces class-discriminative heatmaps by computing gradients of the predicted class score with respect to the final convolutional layer activations [6]. For class c, the importance weight for feature map k is:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{5}$$

The Grad-CAM heatmap is then:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right) \tag{6}$$

These visualizations highlight regions that contribute most to predictions. Fig. 3 shows example Grad-CAM outputs.

V. Experimental Results

A. Classification Performance

The EfficientNetB0 architecture attained 71.32% overall accuracy on the ISIC 2019 evaluation partition. Weighted F1-score reached 0.70, whereas macro-averaged F1-score measured 0.48, reflecting category imbalance influence. The EfficientNet-derived architecture consistently surpassed the foundational CNN across all documented indicators.

Table I presents a comprehensive comparison between the baseline CNN and EfficientNetB0 models across multiple performance metrics.

B. Per-Class Performance Analysis

Fig. 4 presents the detailed classification report showing per-class precision, recall, and F1-score metrics. The model demonstrates strong performance on majority classes (NV: F1=0.85, BCC: F1=0.70) while exhibiting reduced effectiveness on minority classes (DF: F1=0.10, SCC: F1=0.21), highlighting the persistent challenge of class imbalance in dermatological classification tasks.

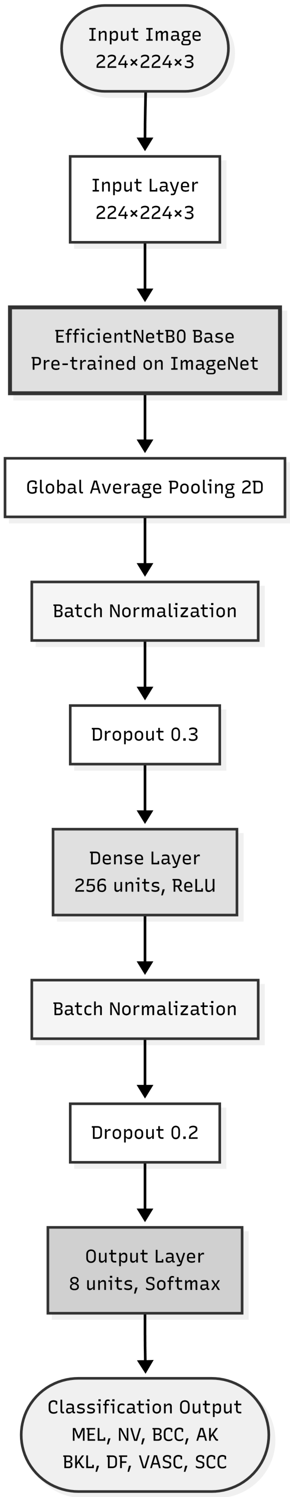

Fig. 2. EfficientNetB0-based architecture for skin lesion classification. Input images (224×224×3) pass through a pretrained EfficientNetB0 backbone, global average pooling, batch normalization, dropout, dense layers, and softmax classification into eight categories.

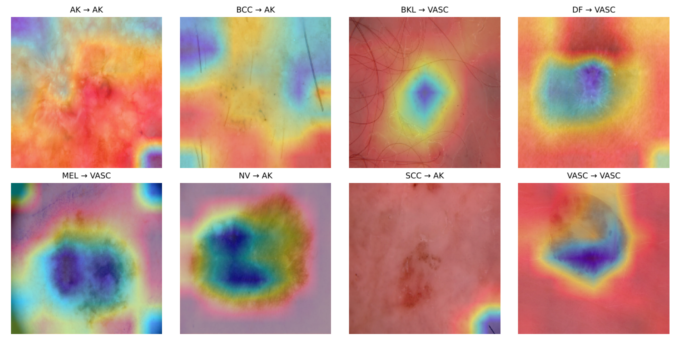

Fig. 3. Grad-CAM heatmaps for ISIC 2019 samples, showing regions most influential for EfficientNetB0 predictions.

TABLE I
PERFORMANCE COMPARISON: BASELINE CNN VS. EFFICIENTNETB0

Metric	**Baseline CNN**	**EfficientNetB0**
Accuracy (%)	58.24	**71.32**
Precision (Weighted)	0.52	**0.70**
Recall (Weighted)	0.58	**0.71**
F1-Score (Weighted)	0.54	**0.70**
F1-Score (Macro)	0.31	**0.48**
Parameters (M)	2.15	4.38
Training Time/Epoch (s)	45	59

CLASSIFICATION REPORT

	precision	recall	f1-score	support
AK	0.4426	0.2061	0.2812	131
BCC	0.6573	0.7535	0.7021	499
BKL	0.4474	0.5508	0.4937	394
DF	0.5000	0.0556	0.1000	36
MEL	0.6642	0.5302	0.5897	679
NV	0.8177	0.8778	0.8467	1932
SCC	0.3889	0.1474	0.2137	95
VASC	0.7000	0.5526	0.6176	38
accuracy			0.7132	3804
macro avg	0.5773	0.4592	0.4806	3804
weighted avg	0.7031	0.7132	0.7007	3804

Fig. 4. Classification report showing precision, recall, and F1-score for each lesion category. The model achieves weighted average metrics of 0.70 for precision, 0.71 for recall, and 0.70 for F1-score across 3,804 test samples.

C. Confusion Matrix Analysis

The classification matrix (Fig. 5) demonstrates robust discrimination capability for predominant categories including NV and BCC. Erroneous classifications principally occur between morphologically similar lesion varieties, particularly MEL and NV, aligning with established diagnostic complexities. The model correctly classified 1,696 of 1,932 NV cases (87.8%) and 376 of 499 BCC cases (75.4%).

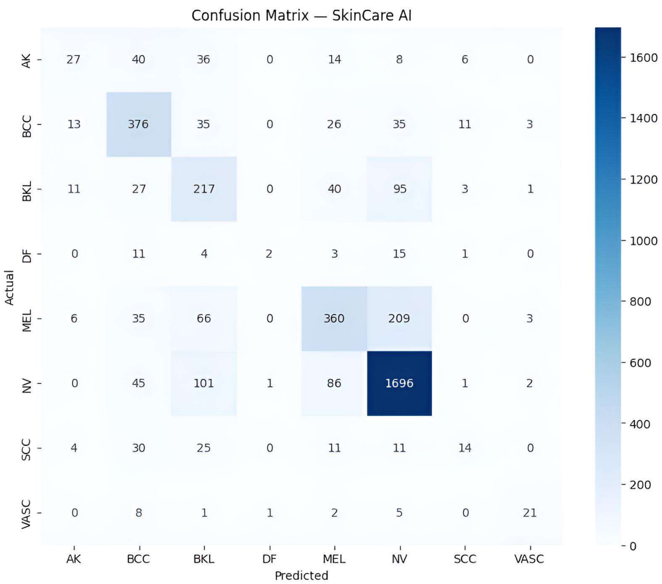

Fig. 5. Classification matrix for EfficientNetB0 evaluated on ISIC 2019, with row-normalized values.

Fig. 6. Training and validation accuracy progression across successive epochs.

D. Training Curves

Training and validation progression curves (Figs. 6 and 7) demonstrate stable convergence with minimal overfitting throughout training epochs. The model achieved best validation accuracy of 72.89% at epoch 7, with consistent improvement across training iterations. Grad-CAM thermal overlays confirm that the architecture concentrates on lesion territories rather than background artifacts, providing qualitative confirmation of clinically meaningful feature acquisition.

VI. System Prototype: SkinCare AI

Trained classifiers integrated into SkinCare AI, a protected internet-accessible investigative prototype. The platform delivers authentication functionality, session-based analysis archives, automated PDF documentation generation, statistical visualization dashboards, and an intelligent conversational

Fig. 7. Training and validation loss progression demonstrating effective generalization.

assistant providing user guidance. All outputs prominently incorporate healthcare disclaimers specifying that this platform serves exclusively academic and investigative purposes rather than clinical diagnostic functions.

VII. Discussion

Experimental outcomes demonstrate that EfficientNetB0 establishes a robust, computationally efficient foundation for multi-category dermatological classification. The model achieved 71.32% accuracy, significantly outperforming the baseline CNN (58.24%), representing a 13.08 percentage point improvement. This performance gain is attributed to the transfer learning paradigm, which leverages pretrained ImageNet features particularly effective for texture and pattern recognition in dermoscopic imagery [18].

Although overall accuracy proves competitive, macro-level indicators highlight persistent difficulties associated with category imbalance and inter-class morphological similarities [14]. The relatively low macro F1-score (0.48) compared to weighted F1-score (0.70) indicates that minority classes such as DF and SCC remain challenging. Future work should explore class-balanced sampling strategies [19], focal loss functions [20], or advanced oversampling techniques such as SMOTE [21].

Grad-CAM integration enhances interpretability, supporting this methodology as a research-oriented analytical support framework. The visualizations confirm that the model attends to clinically relevant lesion features rather than background artifacts, which is essential for building trust in automated dermatological screening systems [22].

Compared to recent state-of-the-art approaches employing ensemble methods [23], our single-model approach offers advantages in terms of computational efficiency and deployment simplicity, making it more suitable for resource-constrained clinical environments [24].

VIII. Conclusion

This manuscript presented an EfficientNetB0-powered framework for multi-category dermatological classification utilizing the ISIC 2019 repository. The proposed methodology achieves dependable performance (71.32% accuracy, 0.70 weighted F1-score) while incorporating interpretable computing techniques enhancing transparency. Comprehensive comparison against a baseline CNN demonstrates significant performance improvements across all evaluation metrics. Outcomes indicate that lightweight, interpretable neural network architectures can function as effective foundational approaches for scalable dermatological analysis within academic investigative contexts. Subsequent investigations will explore sophisticated techniques addressing category imbalance, including focal loss functions and class-balanced sampling, alongside enhanced minority classification performance through targeted augmentation strategies.

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